

2026 IEEE CSR | Conference paper |
Contact: nirab.Hossain@Colorado.edu
moriano@ornl.gov

Evaluating False Alarm and Missing Attacks in CAN IDS

Paper: <https://arxiv.org/pdf/2602.02781>

Code: https://github.com/NirabHossain/ADV_ATTACKS_ROAD_CAN_IVN_IDS



Adversarial robustness of shallow and deep in-vehicle intrusion detection under protocol-compliant, payload-level perturbations

Nirab Hossain

PhD Student

Department of Applied Mathematics

University of Colorado Boulder

Email: nirab.hossain@colorado.edu

Pablo Moriano, PhD

Staff Scientist

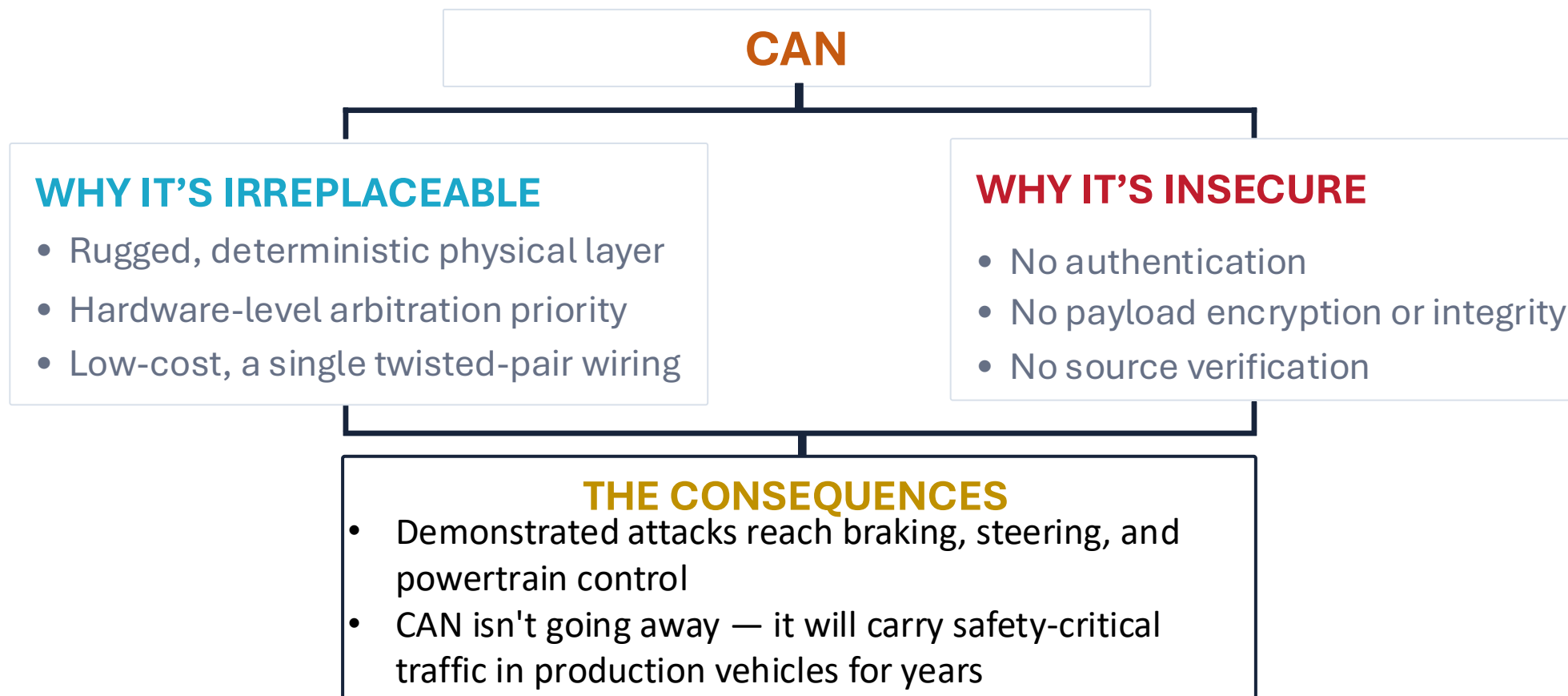
Computer Science and Mathematics Division

Oak Ridge National Laboratory

Email: moriano@ornl.gov

MOTIVATION

A Safety-critical bus net with no security



CAN data frame

SoF

ID

DLC

Data (D0-D7)

CRC

ACK/EoF

THE GAP

Strong on clean data, untested under attack

● Rule / signature

Low overhead, but brittle: cannot generalize to unseen attacks, needs constant updates.



● Shallow ML

DT, RF, ET, XGBoost. Interpretable, cheap to train, strong reported accuracy.



● Deep learning

CNN, LSTM, GNN detectors. State-of-the-art numbers on benchmark CAN datasets.

Prior adversarial work on CAN IDS

Aloraini & Javed '24
False alarms only

Pacheco & Sun '21
Missed attacks only

Zenden et al. '23
FGSM only, no multi-step

Longari et al. '23
Evasion only, oracle-guided

THE SHARED BLIND SPOT

Nearly all of this work reports performance on clean held-out test data. An adversary who already has bus access is not obliged to send clean data and the models are differentiable, or transfer from something that is.

FGSM = Fast Gradient Sign Method
BIM = Basic Iterative Method
PGD = Projected Gradient Descent

Two failure modes, one adversary

An IDS can fail in two directions. Prior work studies them separately. We argue they are **COUPLED**, that is, the same perturbation budget buys either outcome, and a defender has to hold both.



False alarms

Perturb BENIGN frames so the IDS flags them as malicious

Impact: Alert fatigue, unnecessary mitigation, degraded reliability, and erosion of operator trust in the IDS. A denial-of-service on the defense itself.

$$ASR[FP] = \frac{FP_{adv}}{TN_{adv} + FP_{adv}}$$



Missed attacks

Perturb MALICIOUS frames so the IDS lets them through

Impact: The attack executes with the IDS reporting nothing. Spoofed sensor readings, suppressed alerts, unauthorized actuation silently.

$$ASR[FN] = \frac{FN_{adv}}{TP_{adv} + FN_{adv}}$$

Both are computed over adversarially perturbed samples only; they measure attacker success, not baseline error rates.

What we contribute

01

A unified evaluation

False alarms and missed attacks measured both jointly and separately, on the same models, budgets and data; not in separate papers.

02

Protocol-compliant perturbations

Every adversarial frame is a legal CAN frame: fixed ID and DLC, payload bytes clipped to $[0, 255]$ and re-integerized.

03

Shallow vs. deep, head-to-head

DT, RF, ET, XGBoost and a DNN under identical FGSM / BIM / PGD attacks at $\epsilon \in \{1, 5\}$.

04

Realistic attack scenarios

Six ROAD scenarios: fuzzing plus five verified fabrication attacks from a real vehicle.

05

Reproducible

Full pipeline, models and evaluation code released publicly.



A CAN IDS can look excellent on clean data and still be defeated in either direction by an adversary who changes eight bytes.

METHOD

Threat model: a white-box, gray-bus adversary

■ ACCESS

Gray-bus

The adversary can transmit on the CAN bus and impersonate a legitimate ECU (e.g. via a compromised low-privilege node or a diagnostic port).

■ KNOWLEDGE

White-box IDS

Full knowledge of the detector, enabling gradient computation. This is the strongest reasonable assumption and gives an upper bound on attacker capability.

■ CAPABILITY

Payload bytes only

May modify D0–D7 of frames it sends. Cannot change IDs, DLC, timing, or arbitration without being trivially rejected.

■ OBJECTIVE

Either direction

Induce false alarms to erode trust or evade detection to execute an attack silently or both at once.

Why white-box? It removes attacker uncertainty as a confound. If a detector survives a white-box adversary, robustness is a property of the model; not of the attacker's ignorance.

DATASET

ROAD: 3.5 hours of real vehicle CAN traffic

1,551,508

total frames
after preprocessing

49,663

malicious frames
(3.2% of the data)

6

attack scenarios
1 fuzzing
5 fabrication

Why ROAD? Verified, labelled attacks captured on a real vehicle under dynamometer testing; 12 ambient and 33 attack captures.

Split & preprocessing: 70 / 30 stratified split → 1,086,055 train and 465,453 test frames.

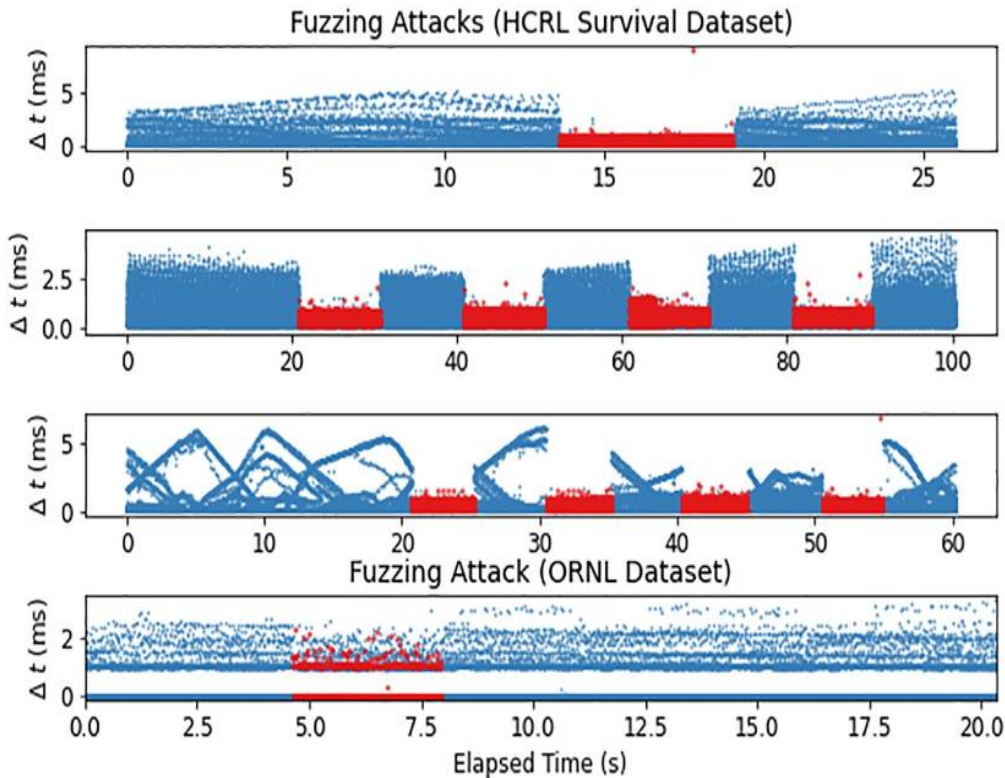
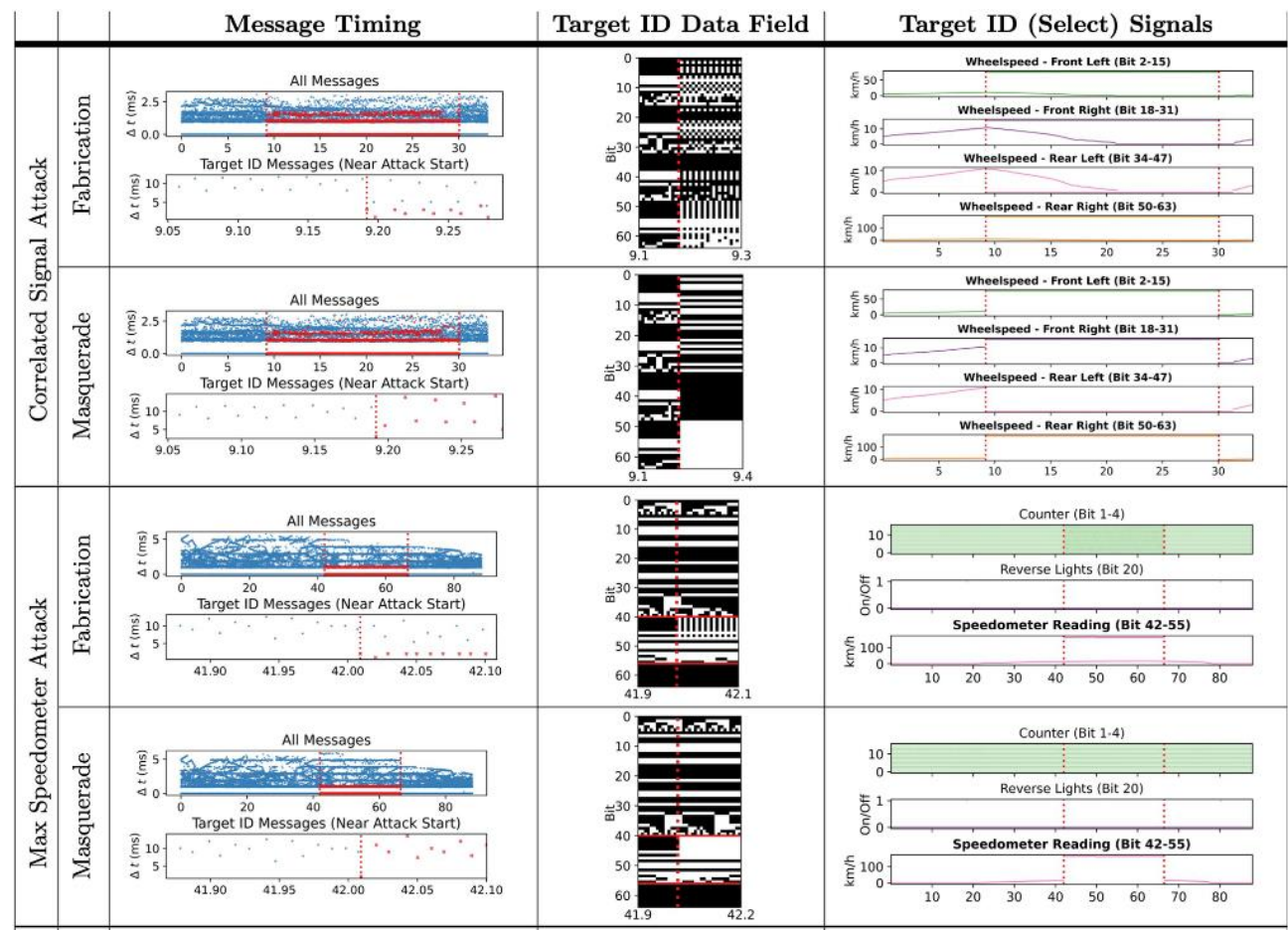
Table I — statistical breakdown of ROAD frames

Attack scenario	Benign	Malicious
FA – fuzzing	87,907	1,061
MECTA Max engine coolant temp	57,932	88
MSA – max speedometer	520,216	17,221
RLOFFA - reverse light off	180,357	10,605
RLONA - reverse light on	480,021	15,195
CSA correlated signal	175,412	5,493
Total	1,501,845	49,663

Dataset: <https://0xsam.com/road/>
Paper: Verma et al | A comprehensive guide to CAN IDS data and introduction of the ROAD dataset
Paper link: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0296879>

DATASET

ROAD dataset signal-level intuition



Source: Verma et al. (2023), "A Comprehensive Guide to CAN IDS Data"-Figs. 3 & 5.

Every attack surgically spoofs one physical signal while leaving the rest of the frame — and the bus's timing — looking legitimate; ROAD's fuzzing attack (right, bottom) is deliberately harder to catch on timing alone than older datasets like HCRL Survival (right, top).

METHOD

Three gradient-based attacks of increasing strength

FGSM: Fast Gradient Sign Method

BIM: Basic Iterative Method

PGD: Projected Gradient Descent

FGSM

Single step

$$x' = x + \epsilon \times \text{sign}(\nabla L)$$

One gradient sign step.
Cheap, fast, and the weakest of the three. Goodfellow et al., 2015.

BIM

Iterative

$$x'_{t+1} = \text{Clip}(x_t + a * \text{sign}(\nabla L))$$

Repeated small steps with clipping after each. Refines FGSM within same budget. Kurakin et al., 2016.

PGD

Iterative + random start

$$x'_{t+1} = \text{Proj}_e(x_t + a * \text{sign}(\nabla L))$$

Random initialization inside the ϵ -ball, projection back after each step. The standard strong first-order adversary. Madry et al., 2018.

BUDGET

$\epsilon \in \{1, 5\}$ per byte, L_∞ -bounded small, valid-range integer changes

TOOLKIT

Adversarial Robustness Toolbox (ART), IBM

TRANSFER

Crafted on a differentiable DNN surrogate, then applied unchanged to DT / RF / ET / XGB

Why MCC, and why ASR over perturbed samples only

Attack success rate (ASR)

$$\text{ASR} = \text{N_success} / \text{N_adversarial}$$

ASR[FP] on benign-only subsets

ASR[FN] on malicious-only subsets

This isolates false-alarm induction from evasion instead of blending them into one error rate.

Matthews correlation coefficient (MCC)

Uses all four confusion-matrix cells.

Ranges -1 (total disagreement) to 0 (random) to +1 (perfect).

Evaluated on the full test set, so it reports joint classification quality — a single scalar that penalizes both failure modes at once.

Source: Chicco and Jurman, BMC Genomics, 2020.

The class-imbalance problem

ROAD is 96.8% benign / 3.2% malicious.

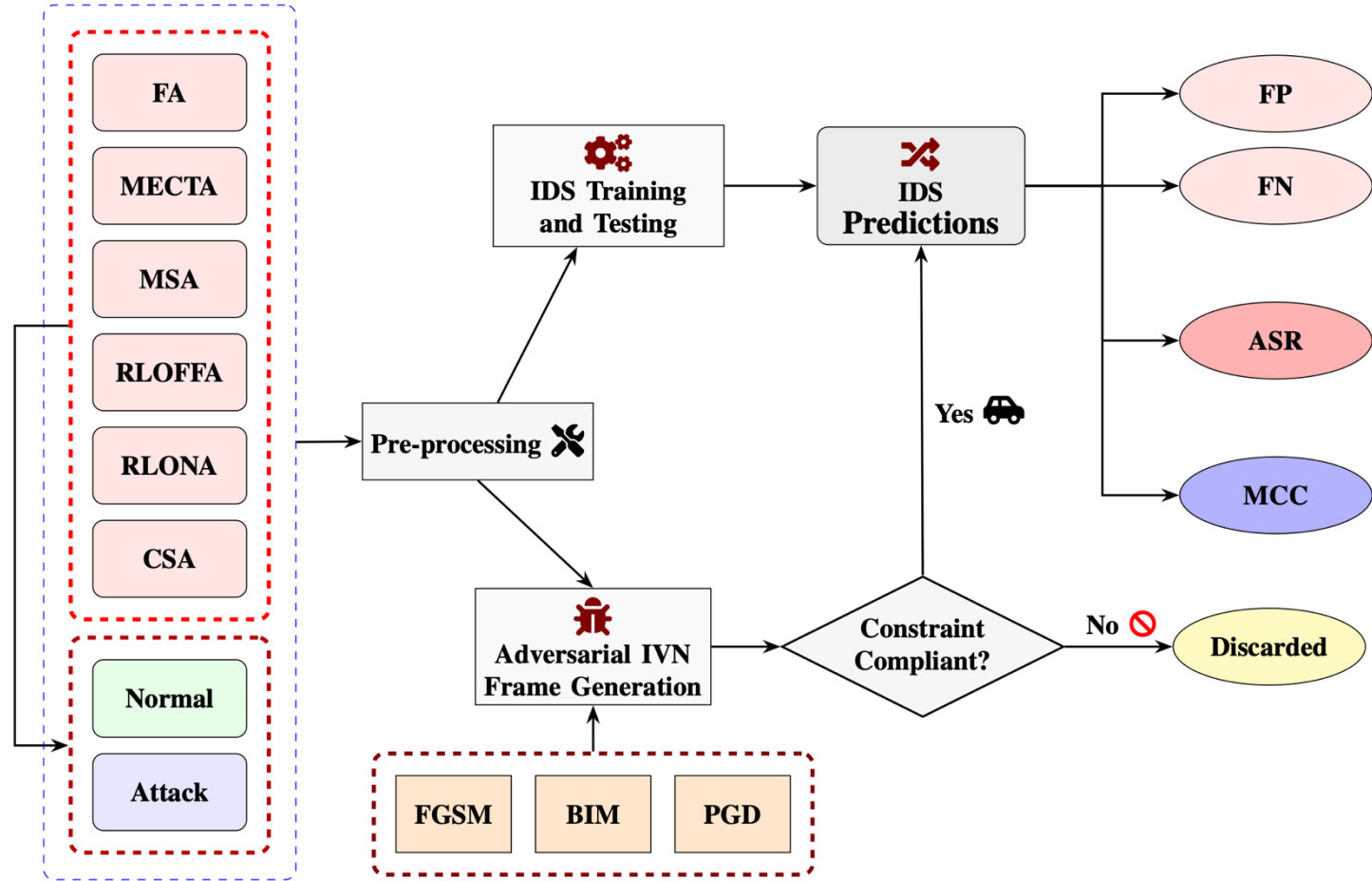
- Under this distribution, accuracy and F1 flatter the majority class.
- A detector that simply never fires would look almost perfect on accuracy.
- MCC does not let that pass: it collapses toward zero whenever either failure mode dominates.

Watch for this in the results:

an MCC at or below zero means the detector is performing no better than random guessing on that scenario. It appears more often than you would expect.

*Exception: CSA, where no false negatives are produced at all, so we report F1
MCC is undefined in the useful sense there.*

Evaluation pipeline



Preprocess six ROAD scenarios → train five IDS → generate FGSM / BIM / PGD perturbations under CAN constraints → discard invalid frames → evaluate FP, FN, ASR and MCC on benign and malicious subsets separately.

RESULTS

Baseline: all five detectors work on clean data

Test set Table: 450,554 benign + 14,899 malicious frames (30% held out).

Model	FP	FN	MCC
DT	941	1,668	0.908
RF	951	1,659	0.908
ET	929	1,683	0.907
XGB	966	1,638	0.908
DNN	384	3,781	0.845

Everything that degrades from here is caused by the attack, not by an inadequate baseline.

Shallow models: MCC $\approx$ 0.91

Near-identical across DT, RF, ET and XGB. Payload bytes alone are highly discriminative on ROAD, so tree ensembles saturate quickly and there is little to choose between them on clean data.

DNN: fewer false alarms, many more misses

384 FP against 3,781 FN — roughly one-third the false alarms of the trees but more than double their misses. The DNN has learned a conservative boundary that favors the benign class, which costs it MCC (0.845) even before any attack.

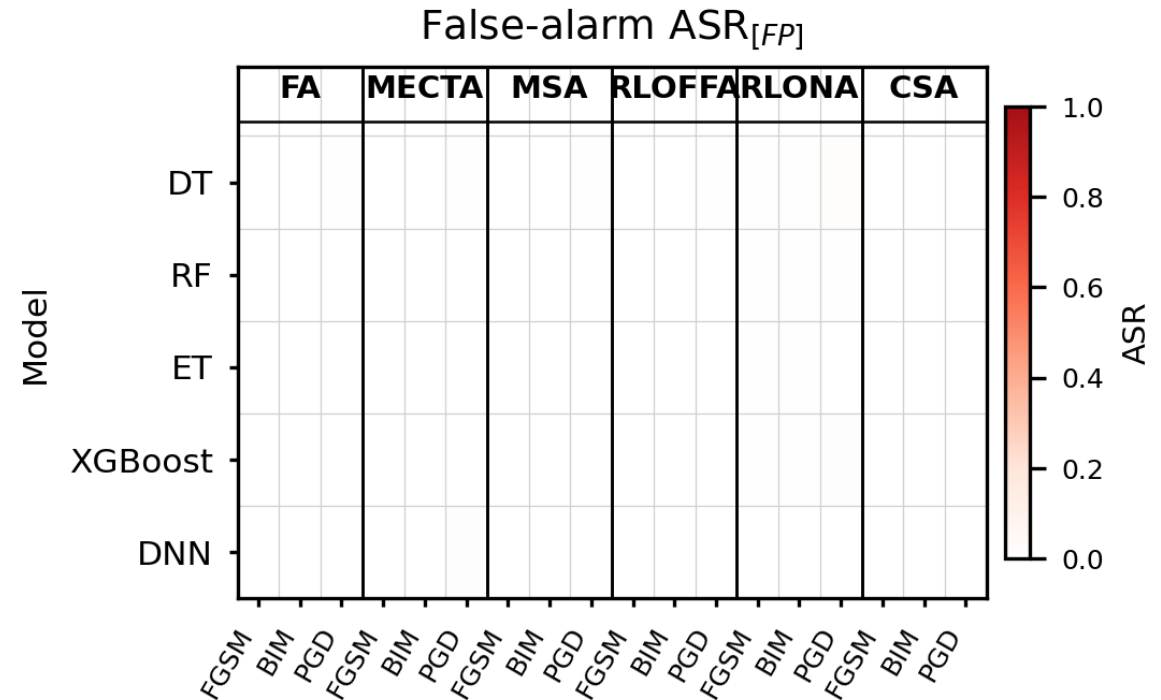
RESULTS

Finding 1 — false alarms are hard to induce

≤ 0.018

Maximum ASR[FP] observed across every model, every attack, every scenario and both ϵ values

- Across all 5 models $\times$ 3 attacks $\times$ 6 scenarios $\times$ 2 budgets, benign frames essentially never flip to malicious.
- The DNN is the strongest here consistently exhibiting the lowest false-alarm ASR of any model, a direct consequence of the benign-leaning boundary from the previous slide.
- Benign MCC stays close to its nominal baseline. The one visible exception is PGD at $\epsilon = 5$, which produces noticeable but not catastrophic degradation.



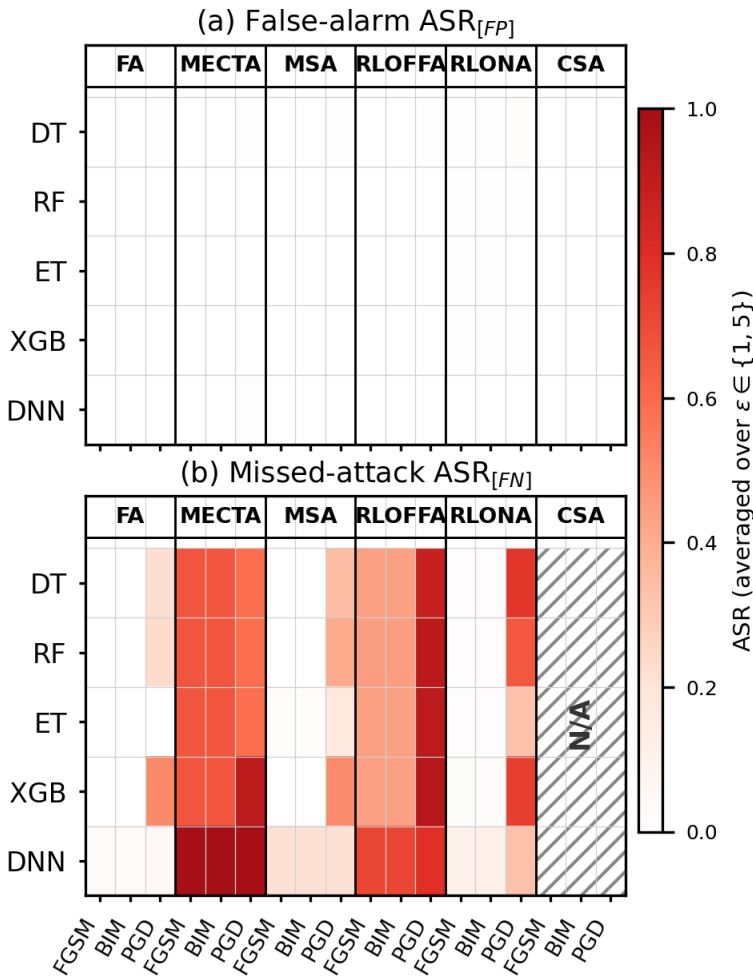
ASR[FP] by model, attack and scenario — averaged over $\epsilon \in \{1, 5\}$. The panel is essentially blank, and that is the finding.

RESULTS

Finding 2 — missed attacks are easy

Same models. Same attacks. Same ϵ .
Same eight bytes. Opposite outcome.

- ASR[FN] = 1.000** The DNN misses every adversarially perturbed **MECTA frame**, under all three attacks. **Complete evasion.**
- MCC ~ 0 on MECTA** All five models collapse to **random-guessing** performance or worse on this scenario.
- ASR[FN] > 0.88** Reached under PGD on RLOFFA by every shallow model — DT 0.88, RF 0.92, ET 0.92, XGB 0.94.
- FA stays safe** Fuzzing is the one scenario where every model holds up; ET reaches ASR[FN] of exactly 0.000.



(a) false alarms · (b) missed attacks

RESULTS

Finding 3 — single-step attacks understate the risk

Mean missed-attack ASR across the five scenarios



FGSM $\approx$ BIM

Identical to three decimal places for every model. Under tight integer clipping, BIM's extra iterations land on the same discrete payload values as FGSM's single step; iteration alone buys nothing here.

PGD changes the picture

Random initialization escapes the single point FGSM and BIM both converge to. XGB goes from 0.23 to 0.72; DT and RF roughly 2.5 $\times$. Only ET and the DNN stay under 0.5.

Takeaway for practitioners: reporting FGSM robustness alone, as several prior CAN IDS studies do, understates achievable evasion by roughly 2 $\times$ on average, and by more than 3 $\times$ for XGB.

RESULTS

Finding 4 — no model wins both directions

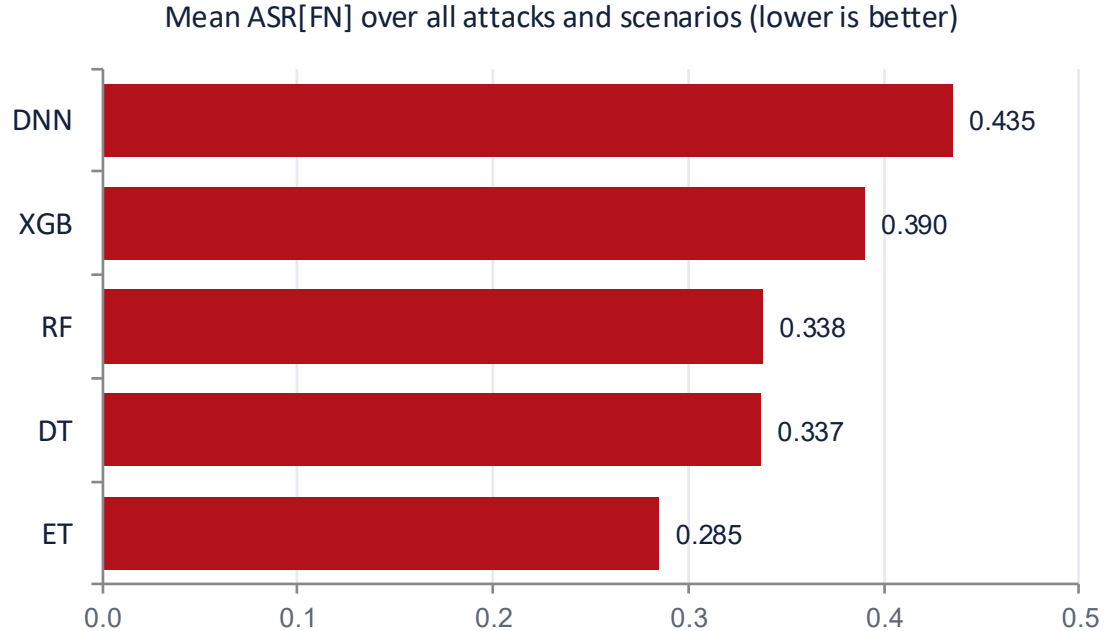


Figure: Mean ASR[FN]

Ensemble randomization reduces, but does not eliminate, adversarial vulnerability in tree-based learners.

ET — best overall trade-off

Lowest mean missed-attack ASR (0.285) and negligible false alarms. Its extreme feature and threshold randomization gives no single consistent gradient direction to exploit.

DNN — best on benign, worst on evasion

Lowest false-alarm ASR of any model, but the highest mean missed-attack ASR (0.435) and the only model reaching complete evasion. The same conservative boundary produces both results.

Caveat: ET's advantage is an average, not a guarantee for LESS DATA. On RLOFFA under PGD it still reaches 0.917 essentially as vulnerable as the rest when *enough data is not available*.

TAKEAWAYS

What a defender should take from this

- 01 Clean-data metrics are not a robustness claim.**
Every model here reports strong nominal performance and several fall to random-guessing MCC under attack. The two rankings do not agree.
- 02 The two failure modes are wildly asymmetric.**
False-alarm ASR never exceeds 2%; missed-attack ASR reaches 100%.
Evasion is the direction that matters the most for CAN IDS, and it is the cheaper one for the attacker.
- 03 Evaluate with a strong first-order adversary.**
FGSM and BIM are indistinguishable under integer clipping. Mean evasion rises from 0.26 to 0.54 under PGD. Single-step-only evaluation is not sufficient evidence.
- 04 Architecture choice is a real robustness lever but not a solution.**
ET is the best trade-off and the DNN the most fragile to evasion, yet every model exceeds 0.88 ASR somewhere. Robustness needs training-time defenses, not just model selection.

Limitations and what comes next

Limitations we own

- We bound perturbation magnitude but do not claim semantic invariance; a perturbed frame may no longer encode its original signal.
- Per-frame classification only. No temporal or sequence-level detectors, which may behave differently.
- One vehicle, one dataset. ROAD is the best available, but generalization across vehicles is untested.
- MECTA carries only 88 malicious frames, so its ASR values are volatile.

Where we go from here

- **Adversarial training**
Retrain under protocol-constrained perturbations and measure the clean-accuracy cost.
- **Input sanitization**
Payload plausibility and range checks upstream of the classifier.
- **Hybrid IDS**
Combine learned representations with interpretable, protocol-aware rules that gradients cannot move.

Thank you

Evaluating False Alarm and Missing Attacks in CAN IDS

- False alarms are hard to induce; ASR[FP] never exceeds 0.018.
- Missed attacks are easy; ASR[FN] reaches 1.000.
- PGD roughly doubles mean evasion over FGSM and BIM and triples it for XGBoost.
- ET is the best trade-off; the DNN is the most fragile to evasion.

Questions

Nirab Hossain

University of Colorado Boulder
nirab.hossain@colorado.edu

Pablo Moriano, PhD

Oak Ridge National Laboratory
moriano@ornl.gov

Paper [arXiv:2602.02781](https://arxiv.org/abs/2602.02781)
Code & data pipeline



This manuscript has been authored by UT-Battelle, LLC under Contract No. DE-AC05-00OR22725 with the U.S. Department of Energy. Research sponsored in part by ORNL's Laboratory Directed Research and Development program.

BACKUP

Supporting materials

22–24 Full per-scenario results (FA, MECTA, MSA, RLOFFA, RLONA, CSA)

25 Metric definitions — ASR and MCC

26 Model configurations

27 Why ET, and where ET fails

28 Positioning against prior work

FPs and FNs are different attacker goals

	Predicted benign	Predicted attack
Actual benign	TN quiet on benign traffic	FP false alarm induction
Actual attack	FN missed attack / evasion	TP detects malicious traffic

- ASR[FP] uses only perturbed benign frames;
- ASR[FN] uses only perturbed malicious frames. Keeping them separate avoids hiding evasion behind false-alarm behavior.
- MCC uses all four confusion-matrix cells and is invariant to which class is called positive.

Full results — FA and MECTA

FA — fuzzing attack

1,061 malicious frames. All values averaged over $\epsilon \in \{1,5\}$.

Model	Benign MCC	FGSM ASR[FN]	BIM ASR[FN]	PGD ASR[FN]	PGD MCC
DT	0.990	0.000	0.000	0.220	0.708
RF	0.989	0.000	0.000	0.232	0.895
ET	0.992	0.000	0.000	0.000	0.998
XGB	0.990	0.000	0.000	0.506	0.820
DNN	0.978	0.038	0.038	0.042	0.990

MECTA — max engine coolant temperature

Only 88 malicious frames — ASR values are volatile.

Model	Benign MCC	FGSM ASR[FN]	BIM ASR[FN]	PGD ASR[FN]	PGD MCC
DT	0.748	0.667	0.667	0.584	0.025
RF	0.748	0.667	0.667	0.584	0.043
ET	0.733	0.667	0.667	0.584	0.045
XGB	0.733	0.667	0.667	0.916	0.000
DNN	-0.000	1.000	1.000	1.000	-0.001

MECTA is the worst scenario for every model: all five reach MCC at or below 0.05 under PGD, and the DNN achieves complete evasion under all three attacks with a benign MCC already at zero — it never learned this scenario in the first place.

Full results — MSA and RLOFFA

MSA — max speedometer attack

17,221 malicious frames — the largest attack subset.

Model	Benign MCC	FGSM ASR[FN]	BIM ASR[FN]	PGD ASR[FN]	PGD MCC
DT	0.962	0.001	0.001	0.347	0.706
RF	0.960	0.002	0.002	0.413	0.712
ET	0.961	0.013	0.013	0.170	0.886
XGB	0.956	0.001	0.001	0.499	0.639
DNN	0.855	0.212	0.212	0.212	0.873

RLOFFA — reverse light off attack

10,605 malicious frames — the most damaging scenario.

Model	Benign MCC	FGSM ASR[FN]	BIM ASR[FN]	PGD ASR[FN]	PGD MCC
DT	0.733	0.438	0.438	0.882	0.159
RF	0.738	0.448	0.448	0.919	0.130
ET	0.733	0.438	0.438	0.917	0.096
XGB	0.742	0.438	0.438	0.938	0.089
DNN	0.612	0.715	0.715	0.790	0.356

RLOFFA is the load-bearing result: a large, well-populated attack subset where every model exceeds ASR[FN] of 0.88 under PGD and MCC collapses below 0.16. Unlike MECTA, this cannot be dismissed as a small-sample artefact.

Full results — RLONA and CSA

RLONA — reverse light on attack

15,195 malicious frames.

Model	Benign MCC	FGSM ASR[FN]	BIM ASR[FN]	PGD ASR[FN]	PGD MCC
DT	0.930	0.018	0.018	0.769	0.035
RF	0.930	0.017	0.017	0.660	0.090
ET	0.930	0.018	0.018	0.325	0.659
XGB	0.930	0.021	0.021	0.735	0.046
DNN	0.797	0.110	0.110	0.331	0.708

RLONA is where ET's advantage is clearest: 0.325 under PGD against 0.66–0.77 for the other tree models, and it retains an MCC of 0.659 while DT, RF and XGB fall below 0.10.

CSA — correlated signal attack

5,493 malicious frames. F1 reported instead of MCC.

No adversarial false negatives were produced for any model or attack.

Every generated perturbation violated a CAN domain constraint after clipping to [0,255] and integer casting, so all candidate frames were discarded at the compliance filter. Benign F1 is 1.000 for the shallow models and 0.999 for the DNN; adversarial F1 stays at 0.998–1.000 throughout, with false-alarm ASR at or below 0.004.

Metric definitions

Attack success rate (ASR)

$$\text{ASR} = \text{N_success} / \text{N_adversarial}$$

$$\text{ASR}[\text{FP}] = \text{FP_adv} / (\text{TN_adv} + \text{FP_adv})$$

$$\text{ASR}[\text{FN}] = \text{FN_adv} / (\text{TP_adv} + \text{FN_adv})$$

Resembles FPR and FNR, but computed exclusively over perturbed samples — so it reads as attacker success, not model error.

Matthews correlation coefficient (MCC)

$$\text{MCC} = \frac{(\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN})}{\sqrt{[(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})]}}$$

-1 = total disagreement · +1 = perfect.
0 = no better than random guessing

Evaluated on the full test set;
FP and FN counts come from benign-only and malicious-only subsets respectively.

Why not F1?

ROAD is 96.8% benign. A trivial always-benign classifier scores 0.968 accuracy and an F1 that looks respectable, while catching nothing. Both metrics are dominated by the majority class.

MCC uses all four confusion-matrix cells and is invariant to which class is called positive. It only reaches high values when the model does well on both classes — which is precisely the property under test when an adversary is pushing in one direction at a time.

Chicco & Jurman, BMC Genomics 2020, make this argument formally.

Exception: CSA produces no adversarial false negatives, so MCC is degenerate there and we report F1 instead.

Model configurations

DT

Decision tree

Single hierarchical rule set over the eight payload bytes. Interpretable, but sensitive to depth — over- or under-fits easily.

RF

Random forest

Bagged randomised trees with random feature subsets at each split. Reduces variance relative to a single tree.

ET

Extra trees

Like RF but split thresholds are also drawn at random, not optimized. This extra randomization is what we credit for its robustness edge.

XGB

XGBoost

Gradient-boosted trees. Captures complex feature interactions across payload bytes; strongest clean-data interaction modelling of the four.

DNN

Deep neural network

Four fully connected hidden layers, 16 ReLU neurons each, single sigmoid output for binary classification. Also serves as the differentiable surrogate for crafting transferred attacks.

Shared setup

All five trained on the same merged 1,086,055-frame training set, evaluated on the same 465,453-frame test set. Features are the eight decimal payload bytes D0–D7 only; no timing, no ID, no sliding-window aggregation. Attacks generated with IBM's Adversarial Robustness Toolbox.

Why ET holds up — and where it does not

The proposed mechanism

ET randomizes both the feature subset and the split threshold at every node, rather than optimizing the threshold. The resulting ensemble has no single consistent decision direction for a transferred gradient to exploit, so perturbations crafted on the DNN surrogate land less reliably.

Where ET fails anyway

RLOFFA under PGD: $ASR[FN] = 0.917$, $MCC = 0.096$. Statistically indistinguishable from DT (0.882), RF (0.919) and XGB (0.938).

On MECTA it sits at 0.584, the same as DT and RF. ET's advantage is an average over scenarios, not a floor.

What we claim

Among the five evaluated models, ET gives the best average trade-off between false-alarm and missed-attack robustness under our threat model.

What we do not claim

That ET is adversarially robust, that it should be deployed on this basis, or that its advantage transfers to other datasets, vehicles or attack constructions.

The general lesson

Ensemble randomization reduces but does not eliminate adversarial vulnerability in tree-based learners. This is consistent with Papernot et al. on transferability.

Positioning against prior work

Study	False alarms	Missed attacks	Multi-step attacks	Protocol constraints	Shallow + deep
Aloraini & Javed, 2024	✓	—	✓	✓	✓
Pacheco & Sun, 2021	—	✓	✓	—	—
Zenden et al., 2023	✓	✓	—	—	—
Longari et al., 2023	—	✓	✓	✓	—
This work	✓	✓	✓	✓	✓

What is genuinely new here

Joint and separate evaluation

Both failure modes on the same models, budgets and data which is what surfaces the asymmetry between them.

Architecture sensitivity

Four shallow families plus a DNN under identical constraints, so differences are attributable to architecture.

Constraint as a finding

The compliance filter is not just methodology — for CSA it removes the entire adversarial sample set.

F1 and MCC answer different questions under imbalance

Why F1 is familiar

F1 summarizes precision and recall for a chosen positive class. It is useful in many IDS papers, but it can hide which side of the confusion matrix failed when the data are highly imbalanced.

Why MCC is the main score here

ROAD is 96.8% benign. MCC uses TP, TN, FP, and FN together, so a detector must do well on both benign and malicious traffic. That is exactly what adversarial robustness tests.

Citation used for the metric choice: Chicco and Jurman, BMC Genomics, 2020.

MOTIVATION

A safety-critical bus with no security

1 No authentication

Any node can transmit any ID. Receivers cannot verify who sent a frame.

2 No encryption or integrity

Payloads are broadcast in the clear to every ECU on the bus.

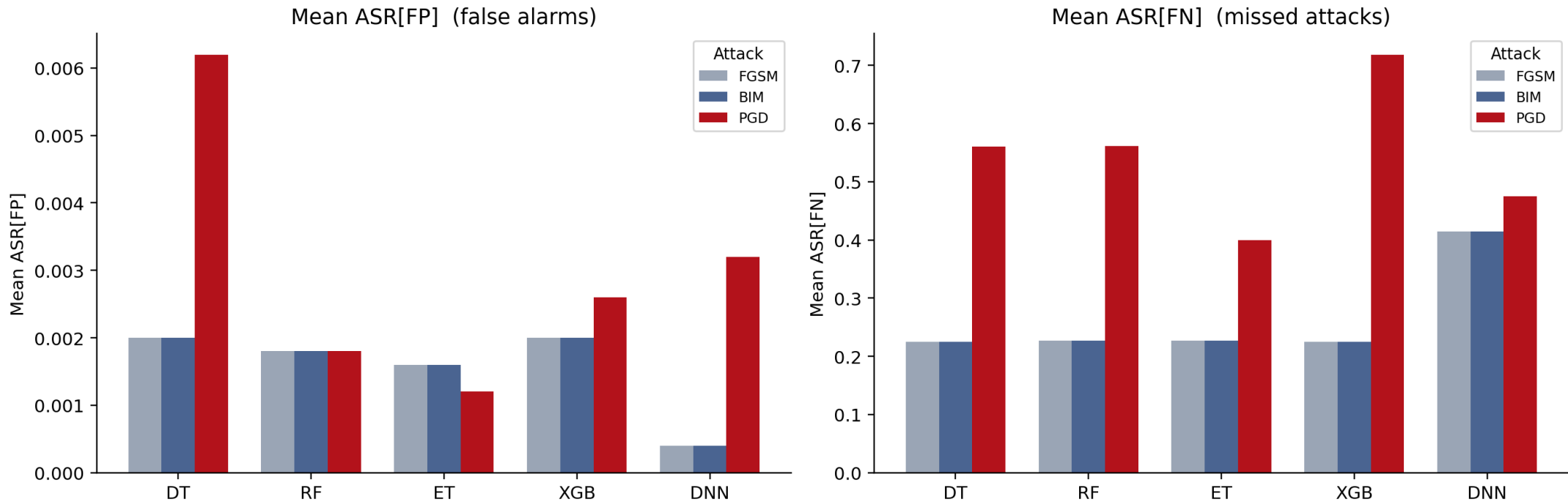
3 Widening attack surface

OBD-II ports, telematics, infotainment and wireless interfaces all reach the bus.

4 Consequences are physical

Demonstrated attacks reach braking, steering, and powertrain control.

False alarms and missed attacks, side by side

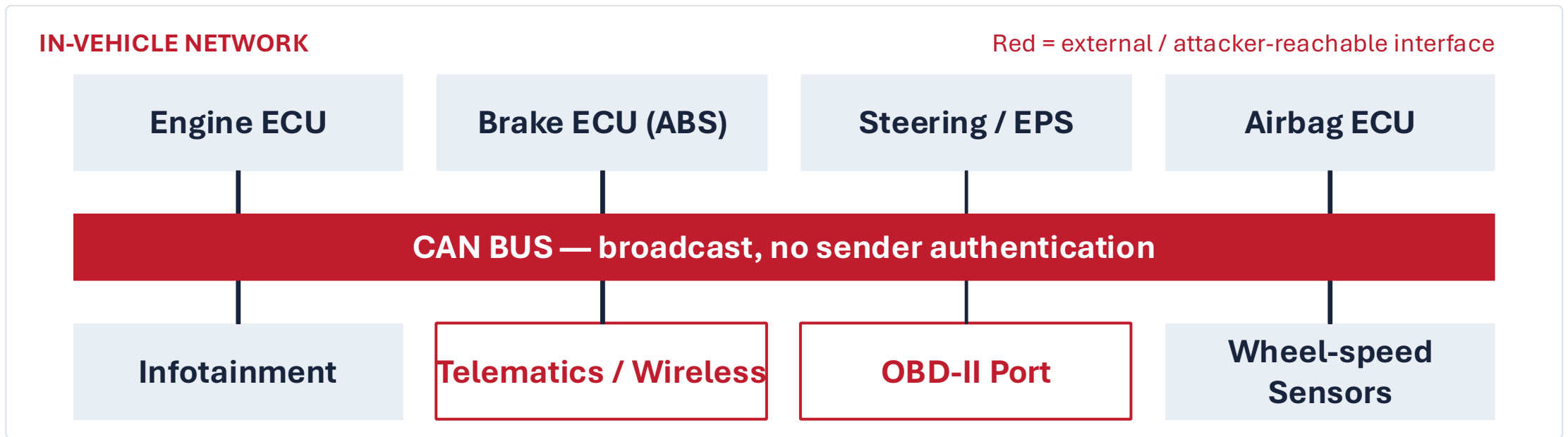


Same budget ($\epsilon \in \{1, 5\}$), same five models — mean ASR[FP] (left) stays near zero across every attack, while mean ASR[FN] (right) climbs sharply under PGD, most visibly for XGBoost. Computed directly from results/tableIII.csv–tableVII.csv.

CONTEXT

One shared bus connects every safety-critical ECU

CAN has connected vehicle electronics since the 1980s and remains the backbone linking engine, brake, steering, and infotainment control today.



METHOD

Constraining the attacker to legal CAN frames

Much AML work in networked systems ignores protocol structure and produces perturbations that are effective in theory and impossible on the wire. We enforce constraints at generation time.

CAN ID *Fixed*

- × ECUs transmit and accept only a manufacturer-configured ID set.

DLC *Fixed*

- × Determined by firmware and hardware.

Timing / arbitration *Untouched*

- × Priority-based arbitration is preserved.

✓ **Mutable: payload bytes D0–D7**

- Gradients are applied only to the eight payload-byte feature columns.
- After every step, values are clipped to the valid byte range [0, 255].
- Values are back to integers before evaluation, i.e., a CAN byte is not a float.
- Any frame that still violates a constraint is discarded rather than sent.

Consequence: an adversarial frame here is a frame that would actually be accepted by the bus. The perturbations are small ($\epsilon \in \{1,5\}$ per byte, L_∞ -bounded) but we make no claim of semantic invariance; we measure detection outcomes, not preserved vehicle semantics.